

SCALAR CLASSES IN CRITICAL HALL MULTIPLICATION

1.1 THE LOWEST COEFFICIENT AT A SCALAR TRIPLE

At a scalar triple, every subspace is invariant. Its Hall coefficient still depends on the ambient cubic potential. Complete flags produce a distinguished class in this coefficient, including at the scalar point. The class determines the lowest-degree images for the potentials of ranks three, four, and five.

All spaces are complex algebraic varieties with their analytic topology. Coefficients are rational. For $1 \leq n \leq 5$, put

$$V_n = \text{Mat}_n(\mathbb{C})^3, \quad f_n(A, B, C) = \text{Tr}(A[B, C]), \quad Z_n = \{[A, B] = [B, C] = [C, A] = 0\}.$$

Differentiation using the trace pairing gives $Z_n = \text{Crit}(f_n)$. Write $\mathfrak{X}_n = [V_n/\text{GL}_n]$ and $\mathfrak{M}_n = [Z_n/\text{GL}_n]$. Constructible complexes on these quotient stacks retain the conjugation action and its descent data. Their ordinary atlas pullbacks have no additional shift. The coefficient is

$$\mathcal{P}_n = \Phi_{f_n} \mathbb{Q}_{\mathfrak{X}_n}[2n^2] \big|_{\mathfrak{M}_n}, \quad \Phi_f K = \text{Cone}(i^* K \longrightarrow \psi_f K)[-1],$$

where $i : f^{-1}(0) \hookrightarrow V$ and ψ_f denotes nearby cycles. This normalization agrees with Massey[1, pp. 1–2]. Thus $H^j(K[a]) = H^{j+a}(K)$ and $\Phi_0 K = K$. The tensor differential is $d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy$. No completion or change of coefficients is used.

For $s = (aI_n, bI_n, cI_n)$, write $P_{n,s}$ for the ordinary atlas stalk of $\mathcal{P}_n$. Translation by scalar matrices preserves f_n , so it suffices to compute $P_{n,0}$. Define the commuting-pair variety and its complement by

$$C_n = \{(A, C) \in \text{Mat}_n(\mathbb{C})^2 : [C, A] = 0\}, \quad U_n = \text{Mat}_n(\mathbb{C})^2 \setminus C_n.$$

Borel–Moore homology below uses locally finite rational chains and the complex orientation of each smooth stratum.

PROPOSITION 1.1.1 (*Scalar stalk and commuting pairs*). For $2 \leq n \leq 5$ and every integer j , there are isomorphisms

$$H^j(P_{n,s}) \simeq \tilde{H}^{j+2n^2-1}(U_n; \mathbb{Q}) \simeq H_{2n^2-j}^{\text{BM}}(C_n; \mathbb{Q}). \quad (1.1)$$

In particular,

$$H^j(P_{n,s}) = 0 \quad (j < -2n), \quad H^{-2n}(P_{n,s}) \simeq \mathbb{Q}.$$

The same last two assertions hold for $n = 1$, where $P_{1,s} = \mathbb{Q}[2]$. Potential monodromy acts as the identity on the entire scalar stalk complex for every rank under consideration.

Proof. Use the positive real ray in the potential plane. By Massey[1, §1, p. 2], the normalized vanishing-cycle stalk is computed by cochains with support in $P = \{\text{Re } f_n \leq 0\}$, restricted to the origin. The fibre

convention is the relative-cochain convention $\text{Fib}(C^*(B) \rightarrow C^*(B \setminus P))$. Both P and its complement are invariant under positive real scaling. The radial map

$$x \mapsto \frac{x}{1 + \|x\|/\delta}$$

maps V_n into the ball of radius δ . Interpolation of the denominator gives homotopy inverses for the inclusions of these pairs. Thus their relative cohomology computes the scalar stalk as δ decreases to zero.

Let $F_n = f_n^{-1}(1)$ and $H = \{z \in \mathbb{C} : \text{Re } z > 0\}$. Choose the cubic root on H that is positive on the positive real axis. Homogeneity gives mutually inverse homeomorphisms

$$V_n \setminus P \longrightarrow F_n \times H, \quad x \mapsto (f_n(x)^{-1/3}x, f_n(x)), \quad (y, z) \mapsto z^{1/3}y.$$

Since V_n and H are contractible, the support complex has cohomology $\tilde{H}^{k-1}(F_n; \mathbb{Q})$ in degree k . This argument fixes the local-to-global comparison at the scalar point.

The cyclic trace identity writes the actual potential as $f_n = \text{Tr}(B[C, A])$. Projection onto (A, C) gives

$$F_n \longrightarrow U_n, \quad (A, B, C) \mapsto (A, C),$$

an affine bundle of complex rank $n^2 - 1$. Indeed, its fibre is the affine hyperplane $\text{Tr}(BD) = 1$, where $D = [C, A] \neq 0$. The trace pairing is nondegenerate. A continuous section is

$$B_0(A, C) = \frac{D^*}{\text{Tr}(D^*D)},$$

where D^* is the conjugate transpose. The affine homotopy from B to B_0 stays in this hyperplane. Therefore projection induces the first isomorphism in (1.1), after the shift $[2n^2]$. Projection is conjugation equivariant. The chosen section only supplies an ordinary topological homotopy inverse.

The real dimension of the pair space is $4n^2$. Its only nonzero Borel–Moore group is $H_{4n^2}^{\text{BM}} = \mathbb{Q}$. The closed-open exact sequence for C_n and U_n , followed by Poincaré duality on the smooth open set U_n , gives

$$\tilde{H}^k(U_n; \mathbb{Q}) \simeq H_{4n^2-k-1}^{\text{BM}}(C_n; \mathbb{Q}).$$

For $k = 0$, the reduced group removes the ambient fundamental class. The complement of a proper complex subvariety in this affine space is connected: a generic complex line joins each point to a common generic intermediate point while avoiding finitely many intersections. The remaining nonpositive degrees vanish. Substitution gives the second isomorphism in (1.1).

For completeness, the top dimension of C_n follows directly from Jordan forms. Fix a Jordan type for A with r distinct eigenvalues and a centralizer of dimension c . Its matrix stratum has dimension $n^2 - c + r$. The matrices C commuting with A form a vector bundle of rank c on this stratum. The corresponding pair stratum has dimension $n^2 + r$. There are finitely many Jordan types. The unique stratum with $r = n$ has dimension $n^2 + n$ and is irreducible: ordered eigenvalues and diagonal entries of C parameterize it, together with a change of basis. The finite permutation quotient preserves irreducibility. Every other stratum has dimension at most $n^2 + n - 1$. Consequently the top Borel–Moore group is

$$H_{2n^2+2n}^{\text{BM}}(C_n; \mathbb{Q}) = \mathbb{Q}, \quad H_m^{\text{BM}}(C_n; \mathbb{Q}) = 0 \quad (m > 2n^2 + 2n).$$

The top generator is the complex fundamental class of the closure of that stratum. No assertion about lower-dimensional components is needed. Equation (1.1) gives the degree bounds.

Finally, at the scalar origin the maps

$$(A, B, C) \mapsto (A, e^{i\theta} B, C), \quad 0 \leq \theta \leq 2\pi,$$

preserve every centered Euclidean ball and multiply f_n by $e^{i\theta}$. They trivialize the Milnor fibres over each circle in the potential plane. At 2π the map is the identity. These trivializations commute with restriction to smaller balls, with the constant-to-nearby-cycle map, and with conjugation. They give identity potential monodromy on the scalar fibre complex. Translation gives the same assertion at every scalar triple. $\square$

The first nonzero groups for the three target potentials are therefore

n	$\dim_{\mathbb{C}} C_n$	first reduced Milnor degree	first degree of $P_{n,s}$
3	12	11	-6
4	20	23	-8
5	30	39	-10.

Each displayed group has rank one. The calculation concerns ordinary atlas stalks. It does not replace equivariant coefficient complexes by these graded vector spaces.

The rank-two scalar fibre gives a further check on the shifts. Write the traceless parts of A, B, C in the basis $\text{diag}(1, -1), e_{12}, e_{21}$ of $\mathfrak{sl}_2$. If T is the matrix of their three coordinate columns, direct expansion gives $f_2 = 2 \det T$. The scalar directions are free. The determinant-one fibre retracts onto $\text{SU}(3)$ by polar decomposition. The fibration $\text{SU}(2) \rightarrow \text{SU}(3) \rightarrow S^5$ has fibre S^3 . Its rational Serre spectral sequence has nonzero terms only in bidegrees $(0, 0), (0, 3), (5, 0), (5, 3)$, so every differential vanishes. The reduced fibre cohomology has degrees 3, 5, 8, each of rank one. After the normalized shift [8], the scalar coefficient has degrees -4, -2, 1.

1.2 COMPLETE FLAGS DEFINE SCALAR GENERATORS

The rank-one coefficient is $\mathbb{Q}[2]$. Complete flags let these coefficients enter the higher-rank critical coefficient without a choice of a scalar Milnor cycle. Let Fl_n be the variety of complete flags in $\mathbb{C}^n$ and set $p_n = n(n-1)/2$. Let $E_n^{\text{fl}} \subset V_n \times \text{Fl}_n$ classify triples preserving a complete flag. Its matrices are upper triangular in an adapted basis, so $f_n|_{E_n^{\text{fl}}} = 0$. The normal bundle of this incidence has complex rank $3p_n$. Its positive complex Thom class followed by flag integration defines

$$\gamma_n^{\text{fl}} : Rq_{n*} \mathbb{Q}_{E_n^{\text{fl}}} \longrightarrow \mathbb{Q}_{V_n}[4p_n],$$

where q_n forgets the flag. The map is proper because Fl_n is projective.

Pass to quotient stacks, and retain the full critical fibre product

$$\mathcal{T}_n = [E_n^{\text{fl}}/\text{GL}_n] \times_{\mathfrak{X}_n} \mathfrak{M}_n, \quad b_n : \mathcal{T}_n \longrightarrow \mathfrak{M}_n.$$

Thus the off-diagonal commutator equations remain part of the source. Proper exchange for vanishing cycles [2, §3, p. 35] and $\Phi_0 = 1$ give

$$\nu_n : Rb_{n*} \mathbb{Q}_{\mathcal{T}_n}[2n] \longrightarrow \mathcal{P}_n, \quad 2n + 4p_n = 2n^2.$$

The unit of pullback and pushforward then defines degree-zero morphisms

$$h_n : \mathbb{Q}_{\mathfrak{M}_n}[2n] \longrightarrow Rb_{n*} \mathbb{Q}_{\mathcal{T}_n}[2n] \xrightarrow{\nu_n} \mathcal{P}_n, \quad \sigma_n = \frac{h_n}{n!}. \quad (1.2)$$

Every construction commutes with changes of frame. Its normal bundles and flag cycles retain their complex orientations on the action diagram. Thus (1.2) defines morphisms on the quotient stacks.

The sign of h_n can be determined at a matrix A with distinct eigenvalues $\lambda_1, \dots, \lambda_n$. In a transverse eigenbasis chart, the actual summand for $i < j$ is

$$(\lambda_i - \lambda_j)(b_{ij}c_{ji} - b_{ji}c_{ij}).$$

Set $\delta_{ij} = \lambda_i - \lambda_j \neq 0$ and introduce

$$\widehat{c}_{ij} = \delta_{ij}c_{ij}, \quad \widehat{c}_{ji} = \delta_{ij}c_{ji}, \quad q = b_{ij}\widehat{c}_{ji} - b_{ji}\widehat{c}_{ij}.$$

Each rescaling is complex linear with real determinant $|\delta_{ij}|^2 > 0$. Both together have determinant $|\delta_{ij}|^4 > 0$, so they preserve the complex normal orientation. The B -polarized positive disk has normalized coordinates

$$(b_{ij}, b_{ji}, \widehat{c}_{ij}, \widehat{c}_{ji}) = (u, v, -\bar{v}, \bar{u}), \quad q = |u|^2 + |v|^2.$$

Its complex parameter orientation defines a positive support generator. Projection to the normalized lower normal coordinates is $(u, v) \mapsto (v, \bar{u})$, of real degree -1 . In the original coordinates it is $(u, v) \mapsto (v, \delta_{ij}^{-1}\bar{u})$. The additional complex multiplication has positive real determinant $|\delta_{ij}|^{-2}$, so the projection degree remains -1 . The lower A normal cancels the flag tangent by multiplication with the nonzero eigenvalue difference. This complex linear map has positive real determinant. If u_n denotes the product of the B -polarized generators, put

$$e_n = (-1)^{p_n} u_n.$$

All support Thom degrees are even. Their ordered external products introduce no sign. Thus each complete flag contributes e_n , and there are $n!$ complete flags. This proves

$$h_n|_{\text{distinct}} = n!e_n, \quad \sigma_n|_{\text{distinct}} = e_n. \quad (1.3)$$

Here $\mathcal{P}_n = \mathbb{Q}[2n]$ on this locus, since its normal quadratic has $2(n^2 - n)$ variables.

THEOREM 1.2.1 (*Extension of the normalized lowest coefficient*). For $1 \leq n \leq 5$, the stalk of σ_n at any scalar triple induces an isomorphism

$$\mathbb{Q} \xrightarrow{\sim} H^{-2n}(\mathcal{P}_{n,s}).$$

Denote the image of 1 by $v_{n,s}$. On the scalar locus, these classes form a constant, conjugation-equivariant cohomology sheaf. They have identity potential monodromy.

Proof. Positive scaling of all three matrices preserves Z_n and each flag incidence, and multiplies f_n by a positive cube. It preserves the support half-plane, the flag cycles and the Thom orientations. Hence the section of $\mathcal{H}^{-2n}(\mathcal{P}_n)$ induced by h_n is equivariant under positive scaling.

Suppose this section had zero stalk at the origin on the atlas. By the definition of a sheaf stalk, it would vanish on a neighborhood of the origin. Every point of Z_n can be scaled into that neighborhood. Scaling equivariance would force the section to vanish everywhere. This contradicts (1.3). Therefore h_n has a nonzero scalar stalk in degree $-2n$. Proposition 1.1.1 shows that this stalk group is one-dimensional. Division by $n!$ gives the stated isomorphism.

Scalar translations preserve the function, incidences and orientations. They transport this generator to every scalar triple. The morphism already descends under conjugation, so its lowest cohomology sheaf has the trivial scalar stabilizer action. The monodromy assertion also follows from Proposition 1.1.1. $\square$

This theorem specifies the lowest cohomology sheaf. It supplies no splitting of the higher cohomology or of the equivariant derived object.

1.3 THE SCALAR IMAGES OF THE HALL COEFFICIENT MAPS

Fix positive integers d, e with $n = d + e \leq 5$. The product order is submodule first, followed by quotient. Let $E_{d,e}$ classify triples preserving a d -plane, and write

$$\mathcal{F}_{d,e} = [E_{d,e}/\mathrm{GL}_n] \times_{\mathfrak{X}_n} \mathfrak{M}_n, \quad \mathfrak{M}_d \times \mathfrak{M}_e \xleftarrow{a_{d,e}} \mathcal{F}_{d,e} \xrightarrow{b_{d,e}} \mathfrak{M}_n.$$

The diagonal block potential is $f_d + f_e$. The normal Thom map has degree $6de$, and Grassmannian integration has degree $-2de$. Their composite is

$$\gamma_{d,e} : Rq_{d,e*} \mathbb{Q}_{E_{d,e}} \longrightarrow \mathbb{Q}_{V_n} [4de].$$

For analytic functions f, g , the relative-cochain product and the support inclusion

$$\{\mathrm{Re} f \leq 0\} \times \{\mathrm{Re} g \leq 0\} \subset \{\mathrm{Re}(f + g) \leq 0\}$$

define the morphism $\Phi_f K \boxtimes \Phi_g L \rightarrow k^* \Phi_{f+g}(K \boxtimes L)$. The product and support inclusion are those of Massey[1, Lemma 1.2, p. 3, and Proposition 1.4, p. 5]. Here k includes the product of zero fibres in the zero fibre of the sum. This morphism is natural in coefficients and proper direct image: both assertions follow from the same maps of relative cochain complexes. For proper direct image, apply proper base change and the proper Künneth isomorphism[2, §1, pp. 13–14]. Its ordered triple products agree because they use the same product cochains and the same inclusion into the support of $f + g + h$.

Use this support morphism, smooth ambient block pullback, proper exchange, and $\Phi(\gamma_{d,e})$. Restriction to the full critical fibre product defines the actual coefficient map

$$\mu_{d,e} : Rb_{d,e*} a_{d,e}^* (\mathcal{P}_d \boxtimes \mathcal{P}_e) \longrightarrow \mathcal{P}_n. \quad (1.4)$$

Its shift identity is $2d^2 + 2e^2 + 4de = 2n^2$. The critical block map need not be smooth. The construction uses the support comparison as a morphism, including for $f_2 + f_3$. Its invertibility is unnecessary here.

Let $\eta_{d,e}$ denote the unit $\mathbb{Q}_{\mathfrak{M}_n} \rightarrow Rb_{d,e*} \mathbb{Q}_{\mathcal{F}_{d,e}}$. The canonical tensor identification and the maps σ_d, σ_e give a morphism from $\mathbb{Q}_{\mathcal{F}_{d,e}} [2n]$ to the incoming coefficient in (1.4).

THEOREM 1.3.1 (*Scalar lowest-degree Hall images*). For $d + e = n \leq 5$, the following equality holds as morphisms of constructible complexes on $\mathfrak{M}_n$:

$$\mu_{d,e} \circ Rb_{d,e*} a_{d,e}^* (\sigma_d \boxtimes \sigma_e) \circ \eta_{d,e} [2n] = \binom{n}{d} \sigma_n. \quad (1.5)$$

At a scalar triple, the source and target of the induced map in degree $-2n$ are both one-dimensional. In the generators specified by $v_{d,s}, v_{e,s}$ and $v_{n,s}$, this map is multiplication by $\binom{n}{d}$. Its image is the whole lowest-degree target and its kernel in that degree is zero.

Proof. Refine an invariant d -plane by a complete flag in that plane and a complete flag in its quotient. The result is exactly a complete flag in $\mathbb{C}^n$. This identifies the refinement square on the full critical stacks, including its frame changes. Proper base change identifies the corresponding iterated direct images.

The forbidden lower entries for the refined flag are the disjoint union of internal entries in the two blocks and the cross entries. The product of their complex Thom classes is the complete-flag Thom class. Integration over the two internal flag varieties and then the Grassmannian evaluates on the same

complete-flag fundamental cycle. The identity $p_n = p_d + p_e + de$ gives the total degree $4p_n$. All these degrees are even.

On coefficients, proper naturality of the support product commutes the internal $\Phi(\gamma_d^{\text{fl}})$ and $\Phi(\gamma_e^{\text{fl}})$ through the external support morphism. Before the internal pushforwards, all one-dimensional block potentials are zero. Their ordered support product is the constant coefficient on the complete-flag incidence. After the pushforwards, the target potentials remain the actual f_d and f_e . The composite therefore equals h_n when its inputs are h_d, h_e . The adjunction units compose to the complete-flag unit because they send the constant section to the same constant section on the refinement. Dividing by $d!e!$ and using $h_n = n!\sigma_n$ proves (1.5).

At a scalar atlas point, proper base change identifies the incoming stalk with hypercohomology on $\text{Gr}(d, n)$ of the pulled-back coefficient family. Proposition 1.1.1 puts that family in degrees at least $-2d - 2e = -2n$. Theorem 1.2.1 identifies its cohomology sheaf in degree $-2n$ with the constant sheaf generated by $v_d \otimes v_e$. The hypercohomology spectral sequence has no term below this degree. Its degree $-2n$ term is $H^0(\text{Gr}(d, n), \mathbb{Q}) = \mathbb{Q}$. It has no incoming or outgoing differential, since the other relevant terms would have negative sheaf-cohomology degree or lower coefficient degree. Equation (1.5) proves the scalar assertion. $\square$

In particular the first images involving f_3, f_4, f_5 are

(d, e)	degree	incoming lowest class	image
$(2, 1)$	-6	$v_2 \otimes v_1$	$3v_3$
$(2, 2)$	-8	$v_2 \otimes v_2$	$6v_4$
$(3, 1)$	-8	$v_3 \otimes v_1$	$4v_4$
$(4, 1)$	-10	$v_4 \otimes v_1$	$5v_5$
$(2, 3)$	-10	$v_2 \otimes v_3$	$10v_5$

Each incoming class denotes the constant section on the corresponding Grassmannian, with the coefficient morphisms just constructed.

1.4 THREE CONSTITUENTS AND THE REMAINING SCALAR COEFFICIENTS

For an ordered triple (d, e, f) of positive ranks with sum $n \leq 5$, let $\mathcal{F}_{d,e,f}$ be the full critical flag stack with these successive dimensions. Its maps a and b remember the constituents and the total object. The ordered triple support product and the flag Thom map define

$$M_{d,e,f} : Rb_* a^*(\mathcal{P}_d \boxtimes \mathcal{P}_e \boxtimes \mathcal{P}_f) \longrightarrow \mathcal{P}_n.$$

Both parenthesizations use this source, by proper base change. Their ambient Gysin degrees add to $4(de + df + ef)$, since

$$de + (d + e)f = ef + d(e + f).$$

Their normal entries partition the same forbidden entries, and their flag integrations evaluate on the same complex cycle. Ordered support associativity and its coefficient naturality therefore give equality of the two derived maps with $M_{d,e,f}$. This argument retains f_{d+e} and f_{e+f} on the intermediate representation spaces. It introduces no permutation of the inputs.

Refinement to complete flags gives the following restriction of this identity to the constructed coefficient morphisms:

$$M_{d,e,f}(\sigma_d, \sigma_e, \sigma_f) = \frac{n!}{d!e!f!} \sigma_n.$$

The left side includes the adjunction unit, as in (1.5). At a scalar point its source in degree $-2n$ is again one-dimensional, by the same hypercohomology argument on the connected flag variety. For the first three scalar comparisons the two factorizations give

(d, e, f)	left factorization	right factorization	image
$(1, 1, 1)$	$2 \cdot 3$	$2 \cdot 3$	$6v_3$
$(2, 1, 1)$	$3 \cdot 4$	$2 \cdot 6$	$12v_4$
$(2, 2, 1)$	$6 \cdot 5$	$3 \cdot 10$	$30v_5$

These are images of scalar coefficient classes. They follow from the complete-flag morphism and its nonzero scalar restriction.

The lowest coefficient has support on the entire scalar locus $\mathbb{C}^3 \times BGL_n$ and on the distinct-eigenvalue locus. The translation and conjugation compatibilities specify the constant lowest cohomology sheaf on the scalar locus. The ambient derived coefficient remains in every incidence and refinement.

Higher scalar images require further calculations. For example, the rank-two scalar coefficient has degrees $-4, -2, 1$. On a scalar $(2, 1)$ incidence, its degree -2 coefficient tensored with the line coefficient starts in total degree -4 . That degree also receives $H^2(\mathbb{P}^2, \mathbb{Q})$ from the lowest coefficient. The morphism from these contributions to $H^{-4}(P_{3,s})$ is not determined by (1.5). It requires the actual coefficient family, its hypercohomology differentials, and its map to the scalar target. Likewise, the odd rank-two class remains part of the $(2, 2)$ and $(2, 2, 1)$ inputs. Equation (1.1) describes the remaining scalar target groups through commuting-pair homology. It does not identify the Hall maps on those groups.

The constructed scalar images and derived flag identities are rational constructible statements through rank five. A comparison with a holomorphic field-theory carrier requires an additional morphism and compatibility with its operations. Tate structures, higher homotopies and further ranks require their own constructions.

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